

ASP.NET MVC Assignment - Product Management System

Objective

Develop a Product Management System using ASP.NET MVC. The application should demonstrate MVC architecture, CRUD operations, Entity Framework, SQL Server integration, validation, and a responsive UI.

Technology Stack

Framework: ASP.NET MVC 5

Language: C#

Database: SQL Server

ORM: Entity Framework 6 (Database First or Code First)

Frontend: Razor Views, HTML5, CSS3, Bootstrap, JavaScript/jQuery

IDE: Visual Studio 2022

Modules

1. Dashboard (Total Products, Categories, Active/Inactive counts)
2. Category Master (Add/Edit/Delete/List/Search)
3. Product Master (Add/Edit/Delete/View)
4. Product List (Search, Filter, Pagination)
5. Product Details

Category Table

Id (PK), CategoryName, Status, CreatedDate

Product Table

Id (PK), CategoryId (FK), ProductName, Description, Price, Quantity, Image, Status, CreatedDate

Functional Requirements

- Add, Edit, Delete Categories
- Add, Edit, Delete Products
- Product image upload (.jpg/.jpeg/.png)
- Category dropdown in Product form
- Search by Product Name
- Filter by Category and Status
- Pagination (10 records/page)
- Product Details page
- Delete confirmation

Validation

- Product Name Required
- Category Required
- Price greater than 0
- Quantity cannot be negative
- Image validation

Expected UI

Dashboard, Category List, Product List, Add/Edit Product Form, Product Details page with Bootstrap responsive design.

Bonus (Optional)

AJAX Delete, Toastr Notifications, Image Preview, Export to Excel, Repository Pattern, Soft Delete

Project Structure

Controllers: ProductController, CategoryController

Models: Product, Category, DbContext

Views: Product (Index/Create/Edit/Details), Category (Index/Create/Edit)

Content, Scripts, Images

Deliverables

1. Visual Studio Solution
2. SQL Database Script
3. Source Code
4. README with setup instructions

Evaluation Criteria

MVC Architecture (20), CRUD (20), Entity Framework (15), SQL (10), Validation (10), UI (10), Code Quality (10), Exception Handling (5).

Time Limit

2-3 Hours